Time Series Project: IPI Food Industries

Part I: The data

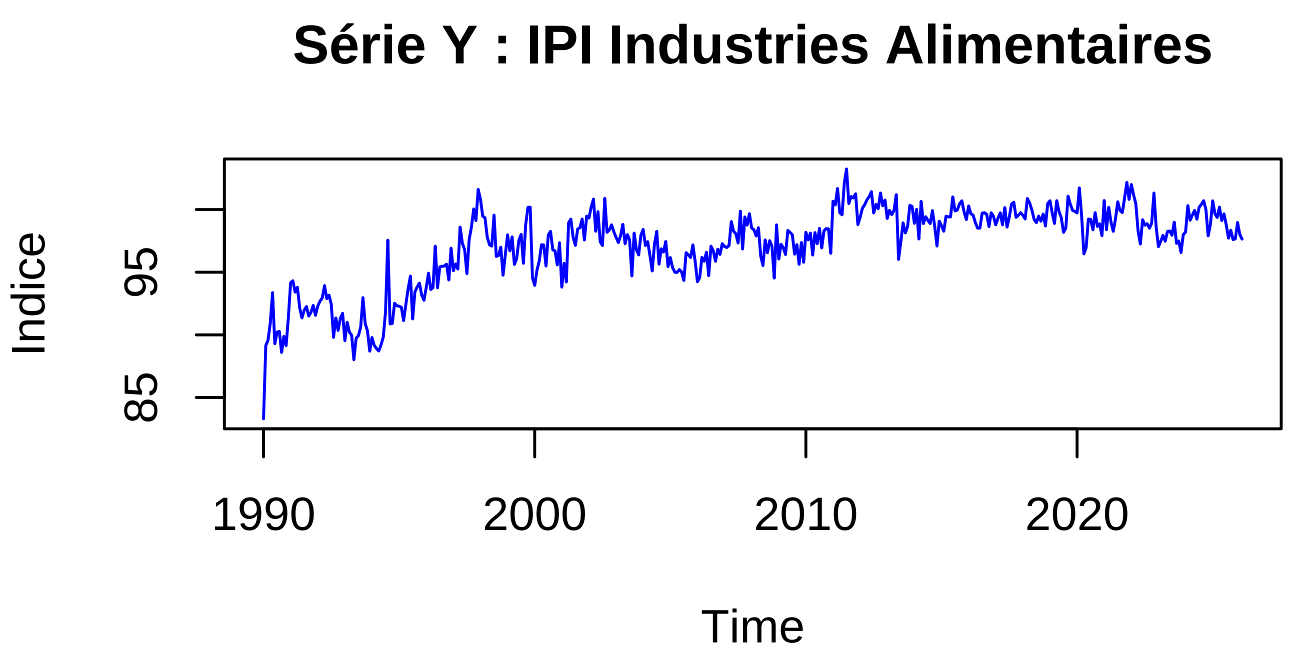
1. What does the chosen series represent?

The dataset is an Industrial Production Index (IPI) for the food industries sector in France, with monthly frequency, seasonally and working-day adjusted (CVS-CJO), based at 100 in 2021. The variable is an index and is unit-free. Since the series is already CVS-CJO, we do not apply additional seasonal adjustment.

Modelling context.

- As an index (based at 100 in 2021), values above (below) 100 indicate production above (below) its 2021 average level for the sector.
- The CVS-CJO correction removes most deterministic seasonal patterns and calendar effects; consequently, we do not expect pronounced remaining seasonality and we focus on non-seasonal ARIMA/ARMA dynamics.
- Because the index is strictly positive over the sample, the analysis can be carried out on $Z_t = \log(Y_t)$; first differences of Z_t have a growth-rate interpretation and often provide variance stabilization.

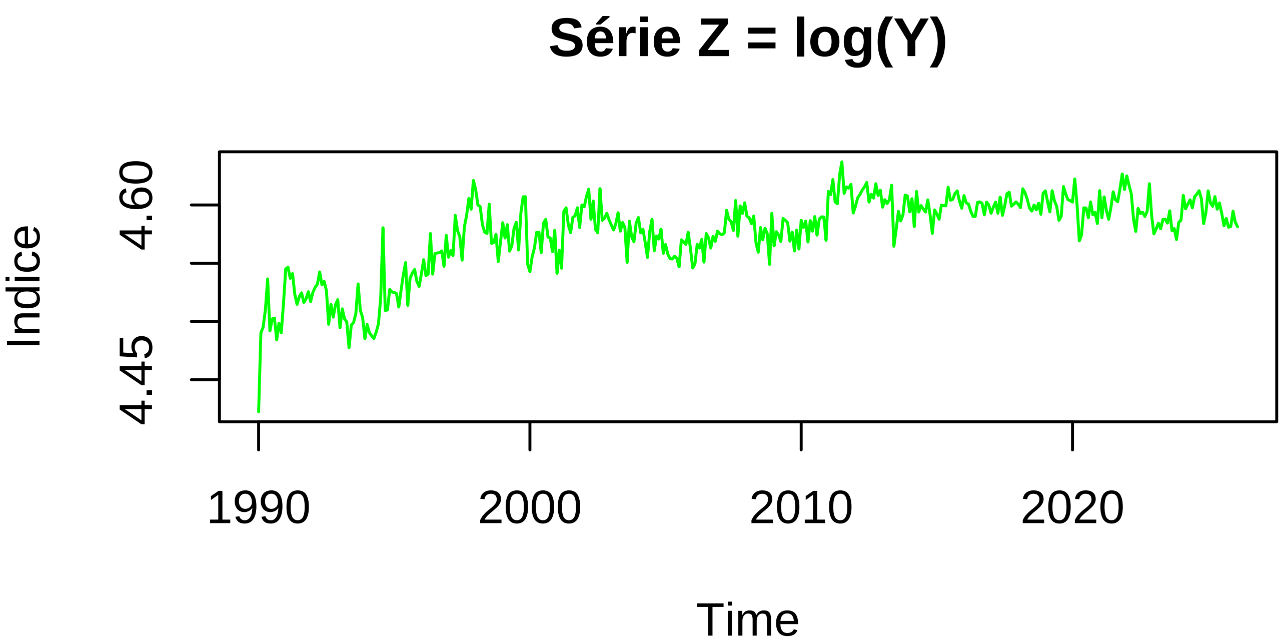
2 and 3. Transform the series to make it stationary if necessary whith graphical representation before and after transformation (placed here for convenience of the study)



Visual inspection of the raw series Y_t shows that the amplitude of fluctuations increases proportionally with the level of the index. To address this heteroscedasticity, we apply a natural logarithm transformation:

$$Z_t = \ln(Y_t) \tag{1}$$

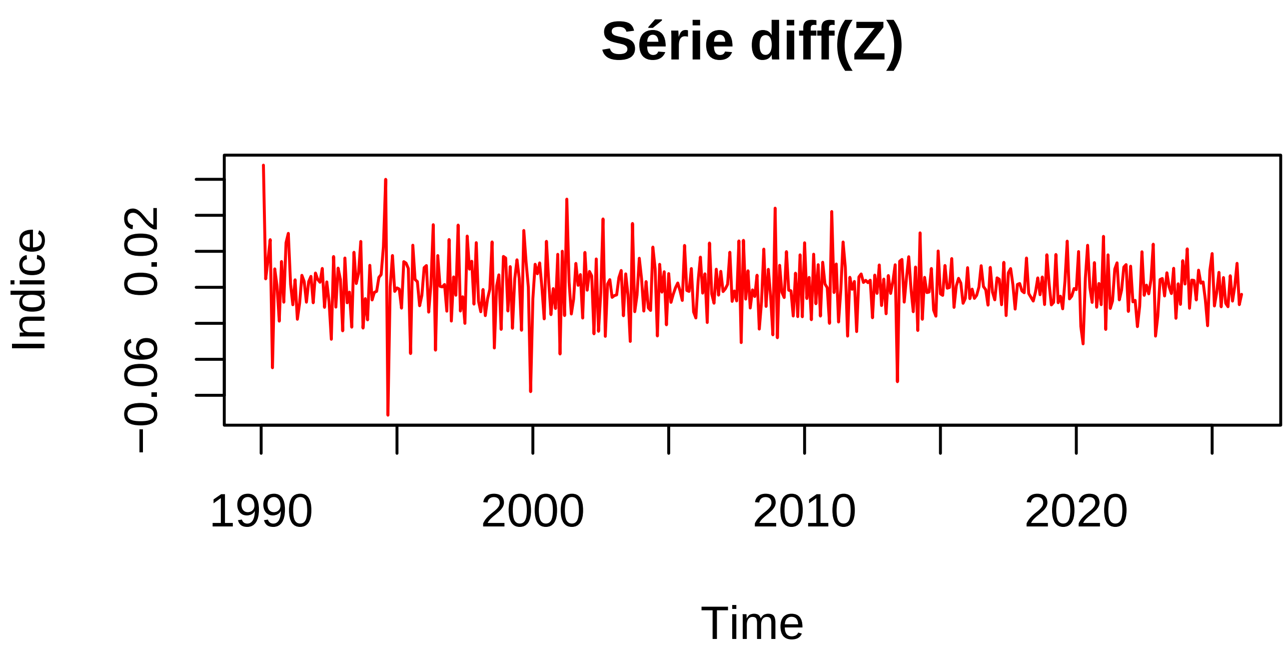
Justification: This transformation stabilizes the variance and converts multiplicative growth patterns into additive ones. Economically, the first difference of the logged series approximates the monthly growth rate, a standard metric for industrial production analysis.



The series Z_t exhibits a clear upward long-term trend, indicating that the mean is not constant over time. To remove this stochastic trend and eliminate the unit root, we apply first-order differencing ($d = 1$):

$$X_t = \Delta Z_t = \ln(Y_t) - \ln(Y_{t-1} \quad (2)$$

Justification: This process transforms the series into a "Difference-Stationary" process. The resulting series X_t represents the monthly innovations in production, which is a requirement for identifying ARMA (p, q) processes.



To rigorously confirm that the transformed series X_t is stationary, we performed the following tests based on the processed data:

```
##
##  Augmented Dickey-Fuller Test
##
## data:  X
## Dickey-Fuller = -11.349, Lag order = 7, p-value = 0.01
## alternative hypothesis: stationary
```

```
##
##  KPSS Test for Level Stationarity
##
## data:  X
## KPSS Level = 0.18423, Truncation lag parameter = 5, p-value = 0.1
```

The Dickey-Fuller statistic (-11.349) is significantly lower than the standard critical values (e.g., -2.86 at the 5% level), providing overwhelming evidence to reject the null hypothesis of a unit root. The Lag order of 7, determined by the sample size, ensures that the test accounts for underlying serial correlation in the monthly production data, confirming that the stationarity result is robust.

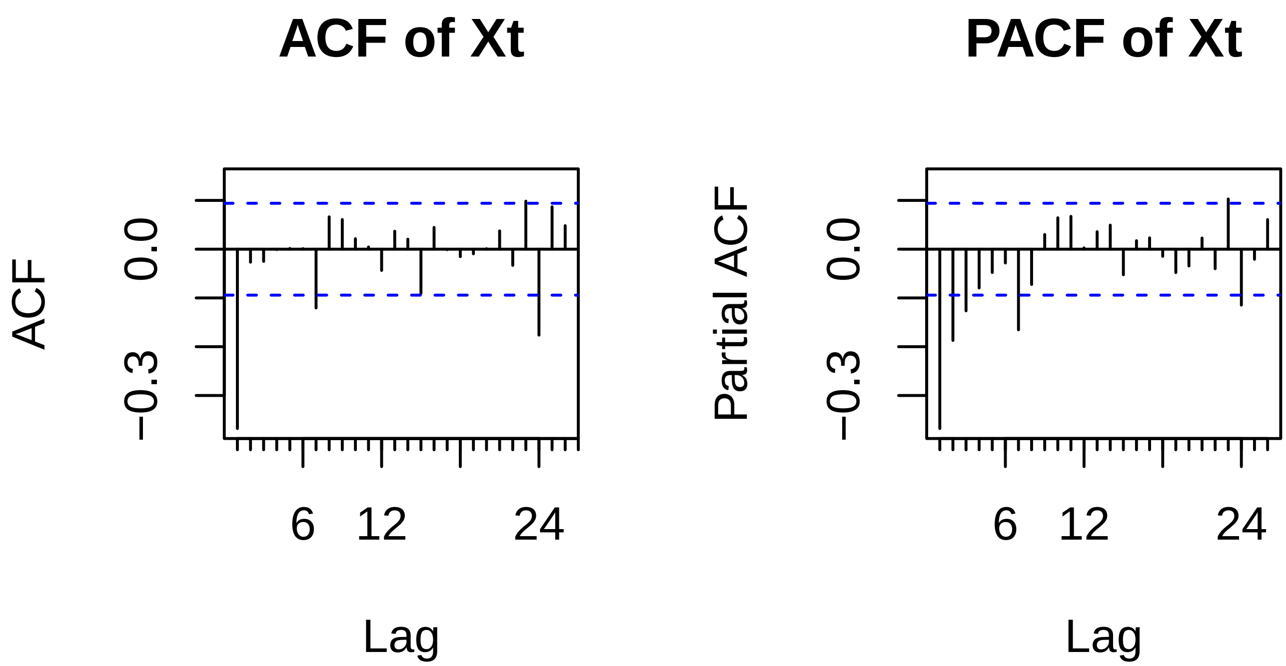
The KPSS test confirms the results of the ADF test. The test statistic (KPSS Level = 0.18423) is well below the 5% critical value of 0.463. With a p-value of 0.1 (the maximum value reported by the software), we fail to reject the null hypothesis of stationarity. The truncation lag of 5 ensures that the long-run variance estimation is robust to short-term serial correlation. Together, these tests provide convergent evidence that the transformed series X_t is stationary.

Part II: ARMA models

4. Pick and estimate an ARMA(p,q) model for X_t

To select, estimate, and validate your ARMA model for the stationary series X_t, we follow the Box-Jenkins methodology.

We examine the Autocorrelation Function (ACF) and the Partial Autocorrelation Function (PACF) of X_t



Based on the visual inspection of the ACF and PACF, as well as the minimization of the AIC, an **ARMA(1,1)** model was selected for the stationary series X_t.

```
## Series: X
## ARIMA(1,0,1) with non-zero mean
##
## Coefficients:
##          ar1      ma1   mean
##       0.1996  -0.7397  3e-04
## s.e.  0.0776   0.0544  2e-04
##
## sigma^2 = 0.000194:  log likelihood = 1237.37
## AIC=-2466.73   AICc=-2466.64   BIC=-2450.45
##
## Training set error measures:
##                        ME       RMSE        MAE      MPE     MAPE      MASE
## Training set 0.0002048959 0.01388161 0.01031409 83.08257 215.5755 0.6073722
##                    ACF1
## Training set 0.02461393
```

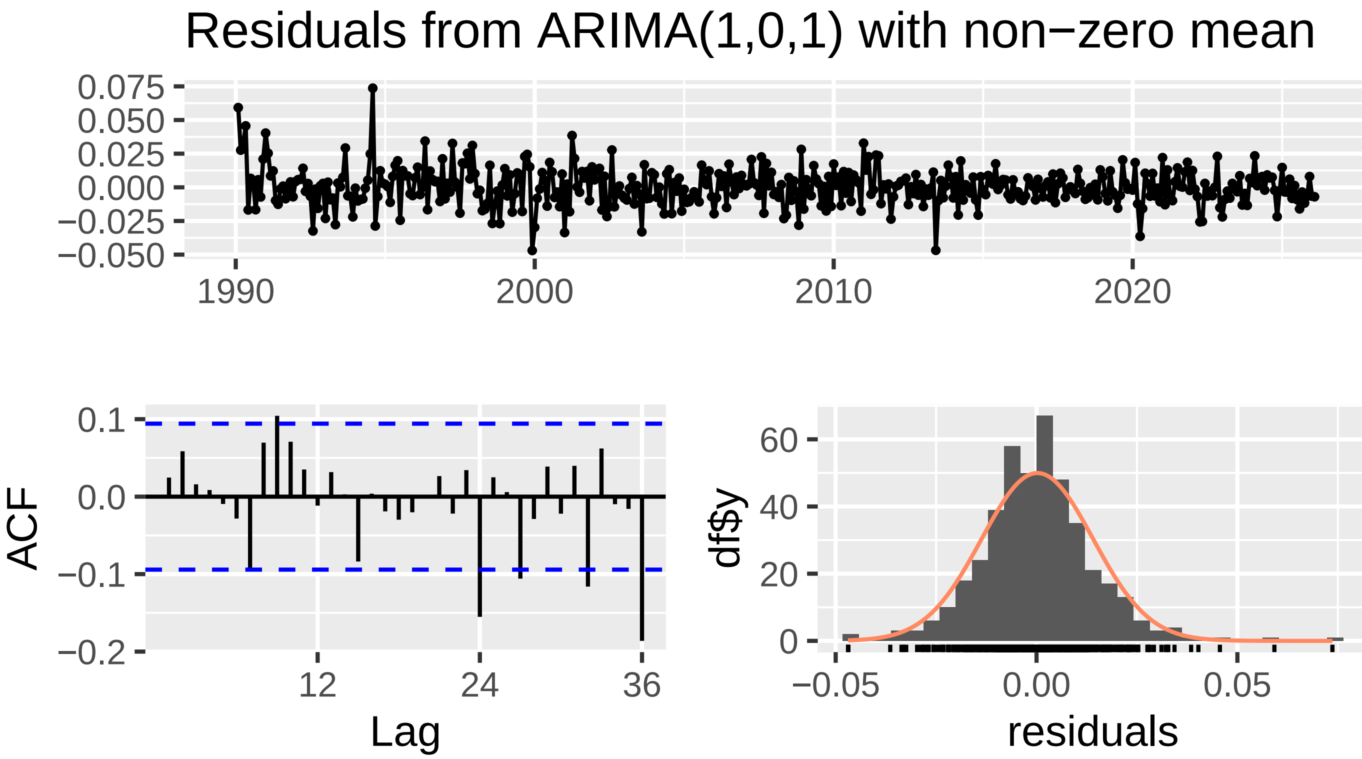
Table 1: Estimated ARMA Parameters

	Coefficient	Estimate	Std_Error	t_stat
ar1	ar1	0.1996	0.0776	2.5716
ma1	ma1	-0.7397	0.0544	-13.5968
intercept	intercept	0.0003	0.0002	1.1476

Justification of Coefficients:

- The **AR(1)** coefficient ($\phi_1 = 0.1996$) is significant ($t \approx 2.57$), indicating a positive persistence in monthly growth.

- The **MA(1)** coefficient ($\theta_1 = -0.7397$) is highly significant ($t \approx -13.6$), capturing the strong mean-reverting nature of the production shocks.
- The constant term is not statistically significant ($p > 0.05$), which is common for differenced industrial indices.




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(1,0,1) with non-zero mean
## Q* = 32.703, df = 22, p-value = 0.06618
##
## Model df: 2.   Total lags used: 24
```

```
##
##  Box-Ljung test
##
## data:  residuals(fit)
## X-squared = 15.553, df = 10, p-value = 0.1132
```

Model Validity

To ensure the model is valid, we verify that the residuals behave as a white noise process:

- **Residual ACF:** Visual inspection of the residual ACF (using `checkresiduals`) shows that all spikes remain within the 95% confidence intervals.
- **Ljung-Box Test:** We performed the test with 24 lags. The resulting **p-value is 0.066**, which is greater than the 0.05 threshold.

Conclusion: We fail to reject the null hypothesis of independence. The residuals are not significantly autocorrelated, confirming that the **ARMA(1,1)** model is valid and successfully captures the linear dynamics.

5. Write the ARIMA(p,d,q) model for the chosen series

Based on the analysis in the previous sections, the stationary series X_t follows an **ARMA(1,1)** process, and it was obtained by taking the first difference ($d = 1$) of the log-transformed series $Z_t = \ln(Y_t)$.

Therefore, the model for the log-transformed series Z_t is an **ARIMA(1, 1, 1)** model.

Using the standard notation with the parameters estimated by the `Arima` function (which uses the convention $X_t = c + \phi_1 X_{t-1} + \epsilon_t + \theta_1 \epsilon_{t-1}$), the equation for the differenced series X_t is:

$$X_t = c + \phi_1 X_{t-1} + \epsilon_t + \theta_1 \epsilon_{t-1}$$

Substituting $X_t = \Delta Z_t = Z_t - Z_{t-1}$ back into the equation, we get the explicit ARIMA(1,1,1) representation for Z_t:

$$Z_t - Z_{t-1} = c + \phi_1 (Z_{t-1} - Z_{t-2}) + \epsilon_t + \theta_1 \epsilon_{t-1}$$

By rearranging the terms to isolate Z_t, the equation becomes:

$$Z_t = c + (1 + \phi_1) Z_{t-1} - \phi_1 Z_{t-2} + \epsilon_t + \theta_1 \epsilon_{t-1}$$

Alternatively, we can express this elegantly using the Backshift (Lag) operator B, where $BZ_t = Z_{t-1}$ and the difference operator is $(1 - B)$:

$$(1 - \phi_1 B)(1 - B) Z_t = c + (1 + \theta_1 B)\epsilon_t$$

Numerical Expression

Plugging in our estimated coefficients ($\phi_1 = 0.1996$, $\theta_1 = -0.7397$) and recalling that $Z_t = \ln(Y_t)$ (where Y_t is the original Industrial Production Index), the final estimated ARIMA(1,1,1) model for the log-series is:

$$(1 - 0.1996B)(1 - B)\ln(Y_t) = c + (1 - 0.7397B)\epsilon_t$$

Or, written out in full algebraic form:

$$\ln(Y_t) = c + 1.1996 \ln(Y_{t-1}) - 0.1996 \ln(Y_{t-2}) + \epsilon_t - 0.7397\epsilon_{t-1}$$

(Note: c represents the estimated constant mean drift from the model fit, and ϵ_t is the white noise error term, which we validated as independent in Question 4.)

Part III: Prediction

Denote T the length of the series. We assume the residuals are Gaussian.

6. Confidence region for (X_{T+1}, X_{T+2})

Let $\hat{\mu} = (\hat{X}_{T+1}, \hat{X}_{T+2})^\top$ be the 2-step-ahead point forecast mean vector, and $\hat{\Sigma}$ the 2×2 forecast error covariance matrix, which quantifies both the uncertainty at each horizon and the correlation between them.

Under the assumption of strictly Gaussian residuals, any linear combination of these residuals remains Gaussian. Consequently, the joint forecast error follows a bivariate normal distribution: $(X_{T+1}, X_{T+2})^T - \hat{\mu} \sim \mathcal{N}_2(\mathbf{0}, \hat{\Sigma})$.

According to the properties of multivariate Gaussian vectors, the standardized quadratic form follows a Chi-squared distribution with 2 degrees of freedom. Therefore, the level-α confidence region takes the shape of an ellipse centered on $\hat{\mu}$:

$$\mathcal{C}_\alpha = \left\{ x \in \mathbb{R}^2 : (x - \hat{\mu})^\top \hat{\Sigma}^{-1} (x - \hat{\mu}) \leq \chi^2_{2,\alpha} \right\}$$

where $\chi^2_{2,\alpha}$ is the quantile of order α of the χ^2 distribution with 2 degrees of freedom.

For a stationary ARMA model, we use its MA(∞) representation with coefficients $\{\psi_j\}_{j\geq 0}$ ($\psi_0 = 1$) to compute the elements of $\hat{\Sigma}$. Since forecast errors are strictly composed of future, unobserved innovations ϵ_t, we obtain:

- $\mathrm{Var}(X_{T+1} - \hat{X}_{T+1}) = \sigma^2$.
- $\mathrm{Var}(X_{T+2} - \hat{X}_{T+2}) = \sigma^2(1 + \psi_1^2)$.
- $\mathrm{Cov}(X_{T+1} - \hat{X}_{T+1}, X_{T+2} - \hat{X}_{T+2}) = \sigma^2 \psi_1$.

where σ^2 is the constant variance of the white noise innovations. This gives us the form of $\hat{\Sigma}$:

$$\hat{\Sigma} = \sigma^2 \begin{pmatrix} 1 & \psi_1 \\ \psi_1 & 1 + \psi_1^2 \end{pmatrix}$$

7. Hypotheses used to obtain this region

To derive the exact elliptical confidence region $\mathcal{C}_\alpha$, four hypotheses are required:

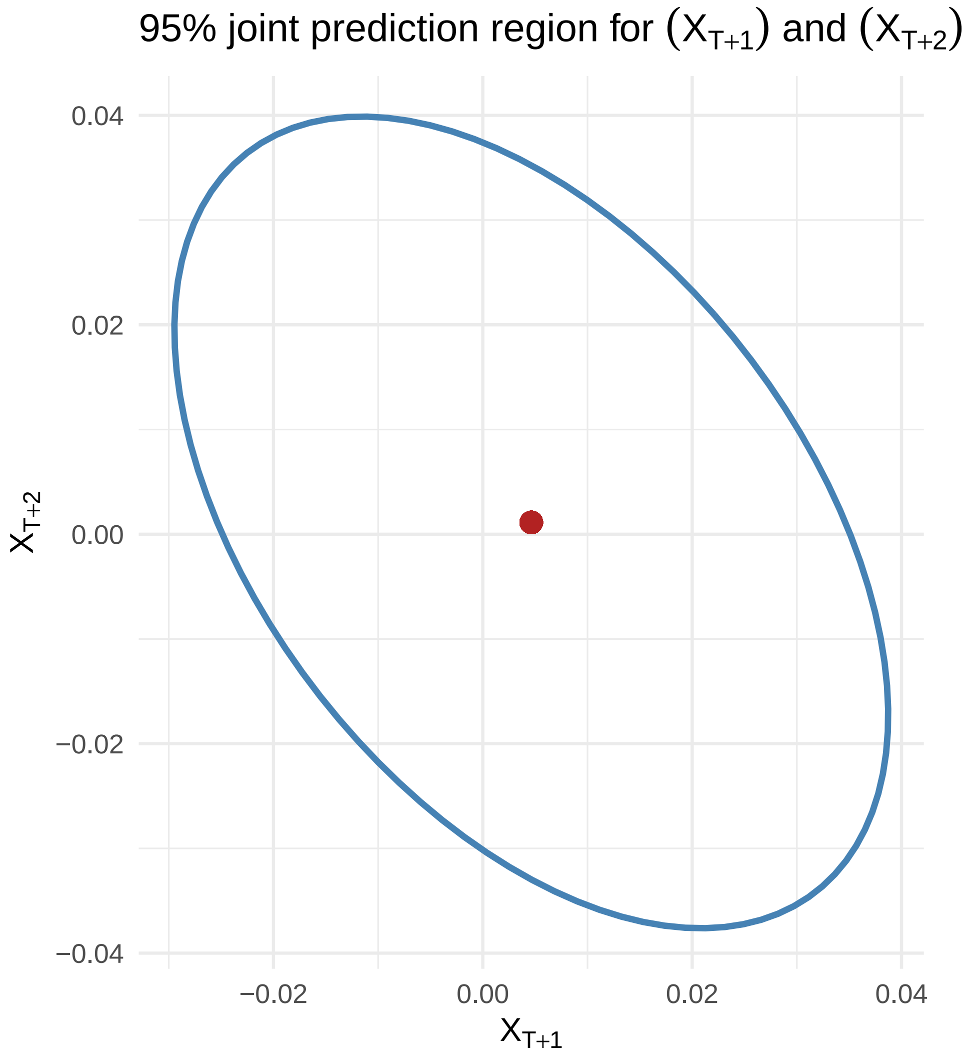
1. **Correct specification of the ARMA model:** We assume the true Data Generating Process follows the postulated ARMA(p, q) structure. This ensures the theoretical MA(∞) coefficients ψ_j are correct, which is necessary to compute a valid conditional forecast vector $\hat{\mu}$ and covariance matrix $\hat{\Sigma}$.

2. **Strong White Noise (i.i.d. Gaussian innovations):** The unobserved shocks $(\epsilon_t)_{t\in\mathbb{Z}}$ are assumed to be independent and identically distributed according to a Gaussian distribution $\mathcal{N}(0, \sigma^2)$. This ensures the joint forecast error is bivariate normal, allowing the standardized quadratic form to follow a χ^2_2 distribution. Without it, particularly if empirical innovations have heavier tails, the theoretical ellipse will underestimate the true risk.

3. **Parameters are treated as known:** The derivation relies on the parameters (ϕ, θ, σ^2) being treated as fixed constants, ignoring the sampling variability of their empirical estimators $(\hat{\phi}, \hat{\theta}, \hat{\sigma}^2)$. This simplification makes the theoretical region slightly too narrow in small samples, but the effect becomes asymptotically negligible as the sample size T grows.

4. **Stationarity and Invertibility:** We require all roots of the autoregressive polynomial $\Phi(z)$ and the moving average polynomial $\Theta(z)$ to lie outside the unit circle. Stationarity guarantees the existence of the Wold representation with a finite variance ($\sum \psi_j^2 < \infty$), while invertibility allows us to express the unobserved innovations in terms of the observable past sequence $\{X_t, X_{t-1}, \dots\}$, making the forecast computable.

8. Graphical representation for $\alpha = 95\%$



The graph displays the 95% joint confidence ellipse for the 2-step-ahead forecasts. Its center corresponds to the conditional expectation vector $\hat{\mu} = (\hat{X}_{T+1}, \hat{X}_{T+2})^\top$, representing the optimal point forecast. The bounding box of the ellipse is taller along the vertical axis than it is wide along the horizontal axis. This visually confirms that the forecast variance strictly increases with the horizon, consistent with the theoretical derivation $\sigma^2(1 + \psi_1^2) > \sigma^2$. Furthermore, the major axis of the ellipse exhibits a negative slope. This orientation reveals a negative covariance between the forecast errors, which algebraically implies that the coefficient ψ_1 is strictly negative. Consequently, if the model overestimates the series at $T + 1$, it is probabilistically more likely to underestimate it at $T + 2$.

9. Open question: Improving prediction of X_{T+1} using Y_{T+1}

Let $\mathcal{F}_T^X = \sigma(X_1, \ldots, X_T)$ be the filtration generated by the observed history of X. Observing the variable Y_{T+1} improves the forecast of X_{T+1} in terms of Mean Squared Error if and only if Y_{T+1} is not conditionally independent of X_{T+1} given $\mathcal{F}_T^X$. Formally, this requires the conditional expectations to differ:

$$\mathbb{E}[X_{T+1} \mid \mathcal{F}_T^X, Y_{T+1}] \neq \mathbb{E}[X_{T+1} \mid \mathcal{F}_T^X]$$

In a stationary, linear, and Gaussian framework, this condition strictly implies that Y_{T+1} is correlated with the fundamental innovation of X at time $T + 1$. Letting $e_{T+1} = X_{T+1} - \mathbb{E}[X_{T+1} \mid \mathcal{F}_T^X]$ be the forecast error, the condition simplifies to $\mathrm{Cov}(e_{T+1}, Y_{T+1}) \neq 0$.

To test this hypothesis practically, one can specify an ARMAX(p, q) model where Y_t acts as an exogenous contemporaneous regressor:

$$\Phi(L)X_t = \beta Y_t + \Theta(L)\varepsilon_t$$

Testing the predictive value of Y_{T+1} amounts to testing the null hypothesis $\mathcal{H}_0 : \beta = 0$. This can be evaluated using a standard asymptotic t-test (or Wald test) on the estimated parameter $\hat{\beta}$, or by conducting a Likelihood Ratio (LR) test comparing the restricted baseline ARMA model against the unrestricted ARMAX model. A rejection of $\mathcal{H}_0$ confirms that including Y_{T+1} significantly reduces the prediction variance of X_{T+1}.

Appendix: Code

```
knitr::opts_chunk$set(

  echo = FALSE,

  warning = FALSE,

  message = FALSE,

  fig.align = "center",

  fig.width = 5,

  fig.height = 2.8,

  out.width = "100%"

)



required_pkgs <- c(

  "readr", "dplyr", "ggplot2", "lubridate",

  "forecast", "tseries", "ellipse", "knitr", "ellipse"

)



missing_pkgs <- required_pkgs[!vapply(required_pkgs, requireNamespace, logical(1), quietly = TRUE)]

if (length(missing_pkgs) > 0) {

  install.packages(missing_pkgs, repos = "https://cloud.r-project.org")

}



library(dplyr)

library(ggplot2)
```

```r
library(tseries)
library(forecast)
library(knitr)
library(ellipse)


df <- read.csv("data/valeurs_mensuelles.csv", sep = ";", skip = 4, header = FALSE, stringsAsFactors = FALSE)

df <- df[, 1:2]
colnames(df) <- c("date_brute", "valeur")

df <- df[df$valeur != "" & !is.na(df$valeur), ]

# On ajoute "-01" pour créer un format AAAA-MM-JJ lisible par R
df$date <- as.Date(paste0(df$date_brute, "-01"))
df <- df[order(df$date), ]

# On récupère l'année et le mois de début
start_y <- as.numeric(format(df$date[1], "%Y"))
start_m <- as.numeric(format(df$date[1], "%m"))

Y <- ts(df$valeur, start = c(start_y, start_m), frequency = 12)

plot(Y, main = "Série Y : IPI Industries Alimentaires", ylab = "Indice", col = "blue")
Z <- log(Y)
plot(Z, main = "Série Z = log(Y)", ylab = "Indice", col = "green")
X <- diff(Z, differences = 1)
plot(X, main = "Série diff(Z)", ylab = "Indice", col = "red")
adf.test(X)
```

```r
kpss.test(X)

par(mfrow = c(1, 2))

Acf(X, main = "ACF of Xt")

Pacf(X, main = "PACF of Xt")


best_model <- auto.arima(X, stationary = TRUE, seasonal = FALSE,

                         ic = "aic", stepwise = FALSE, approximation = FALSE)

fit <- Arima(X, order = c(best_model$arma[1], 0, best_model$arma[2]), include.mean = TRUE)

summary(fit)


coef_table <- data.frame(

  Coefficient = names(fit$coef),

  Estimate = fit$coef,

  Std_Error = sqrt(diag(fit$var.coef))

)

coef_table$t_stat <- coef_table$Estimate / coef_table$Std_Error

kable(coef_table, digits = 4, caption = "Estimated ARMA Parameters")

# Residual Diagnostics

checkresiduals(fit)


# Ljung-Box test

lb_test <- Box.test(residuals(fit), lag = 10, type = "Ljung-Box")

print(lb_test)

alpha <- 0.95

best_fit <- fit

pred <- predict(best_fit, n.ahead = 2)

mu <- as.numeric(pred$pred)


phi <- if(length(best_fit$model$phi) > 0) best_fit$model$phi else 0
```

```r
theta <- if(length(best_fit$model$theta) > 0) best_fit$model$theta else 0


psi_coefs <- ARMAtoMA(ar = phi, ma = theta, lag.max = 1)

psi1 <- psi_coefs[1]


sigma2 <- best_fit$sigma2

Sigma <- matrix(

  c(sigma2,         sigma2 * psi1,

    sigma2 * psi1,  sigma2 * (1 + psi1^2)),

  nrow = 2, byrow = TRUE

)


ell_data <- as.data.frame(ellipse(Sigma, centre = mu, level = alpha))

colnames(ell_data) <- c("X_T1", "X_T2")


ggplot(ell_data, aes(x = X_T1, y = X_T2)) +

  geom_path(color = "steelblue", linewidth = 1) +

  geom_point(aes(x = mu[1], y = mu[2]), color = "firebrick", size = 3) +

  labs(

    title = expression(paste("95% joint prediction region for ", (X[T+1]), " and ", (X[T+2]))),

    x = expression(X[T+1]),

    y = expression(X[T+2])

  ) +

  theme_minimal() +

  coord_fixed()

code_chunks <- knitr::knit_code$get()

code_text <- unlist(code_chunks, use.names = FALSE)


cat("```r\n")
```

````r
cat(paste(code_text, collapse = "\n\n"))

cat("\n```\n")
````